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22024

Biostatistics

Standard Deviation:

Standard deviation tells us how much the data values deviate from the mean (average) of the dataset. In biostatistics, it is used to measure the spread of biological data such as height, weight, blood pressure etc.

Symbol:

Population: σ (sigma)

Sample: s

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Defination: Is defined as

“Positive square root of the mean of squared of data from its mean”

Formula:

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

or

$$\sigma = \sqrt{\left[\frac{\sum X^2}{N} - \left(\frac{\sum X}{N} \right)^2 \right]}$$

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Example:**30, 37, 40, 26, 45, 50**

X	$x - \bar{x}$	$(x - \bar{x})^2$
30	$30 - 38 = -8$	64
37	$37 - 38 = -1$	1
40	$40 - 38 = 2$	4
26	$26 - 38 = -12$	144
45	$45 - 38 = 7$	49
50	$50 - 38 = 12$	144
$\sum X = 228$	$\sum (X - \bar{X}) =$	$\sum (X - \bar{X})^2 = 406$

Mean: $\bar{X} = \frac{\sum X}{N} = \frac{228}{6} = 38$

S.D: $S^2 = \frac{\sum (X - \bar{X})^2}{N - 1}$

$$= \frac{406}{6 - 1} = \frac{406}{5} = 81.2$$

S: $\sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}} = \sqrt{81.2} = 9.01$

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$$S.E.: \quad SX^2 = \frac{S}{\sqrt{n}} = \frac{9.01}{\sqrt{6}}$$

$$SX^2 = 3.68$$

Coefficient of Variation.

$$C.V = \left(\frac{S}{M} \times 100 \right) \%$$

$$C.V = \left(\frac{9.01}{3.8} \times 100 \right) \%$$

$$C.V = 23.71\%$$